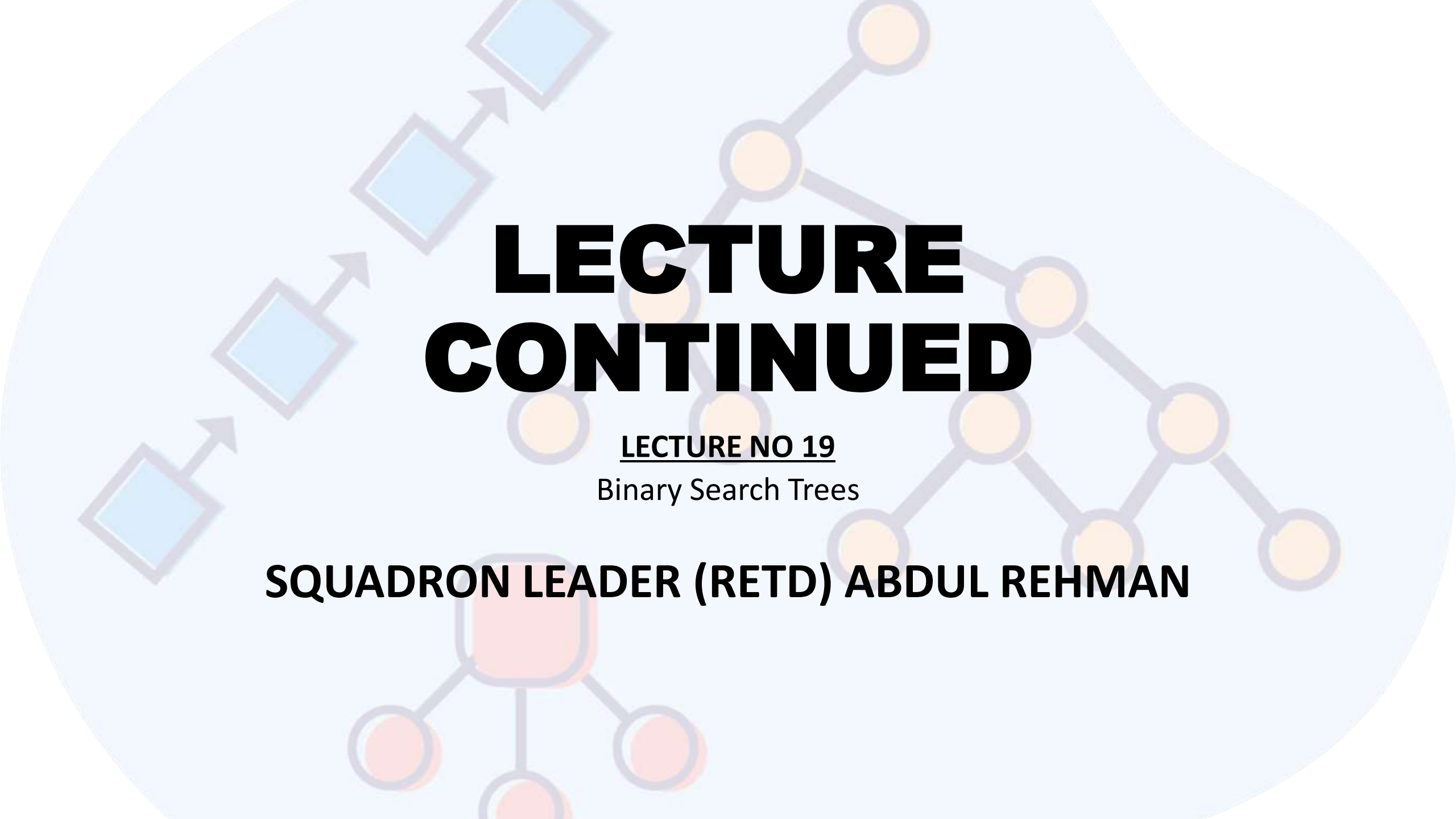




LECTURE CONTINUED

LECTURE NO 19

Binary Search Trees

SQUADRON LEADER (RETD) ABDUL REHMAN

BINARY SEARCH TREES

- A **binary search tree** is a binary tree that:
- Maintains a specific order among its elements to facilitate
 - Efficient search
 - Insertion
 - Deletion
- Each node in a BST follows
 - Left child's value is less than its own value
 - Right child's value is greater than or equal to its own

PROPERTIES OF BST

- **Ordering Property:** For any node in a BST, all keys in the left subtree are less than the key of node, and all keys in the right subtree are greater than or equal to the node
- **Unique Path:** There is a unique path from the root to any other node.
- **Efficient Search:** The ordering property allows for efficient searching, with average and worst-case time complexities of $O(\log n)$ and $O(n)$, respectively, depending on the tree's balance.

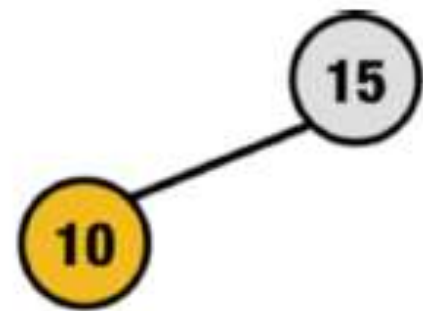
BASIC OPERATIONS IN BST

- **Insertion:** Inserts new elements while maintaining the BST's ordered property. The insertion finds the correct position for the new node to keep the tree sorted.
- **Deletion:** Handles three scenarios: deleting a leaf node, a node with a single child, or a node with two children. Each case is managed to preserve the BST's ordered structure.
- **Searching:** Involves comparing the target value with the nodes, starting from the root and traversing the tree accordingly. This process repeats until the target is found or the search reaches a leaf node.

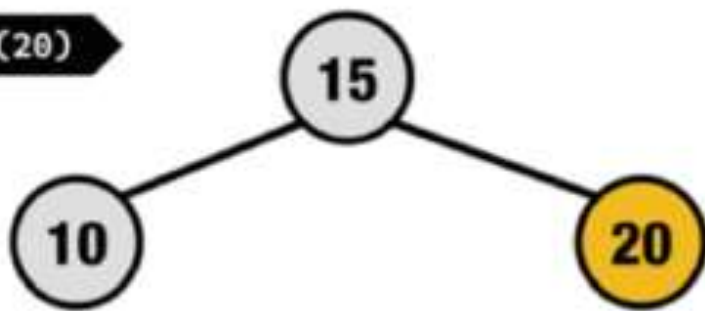
Insert(15)



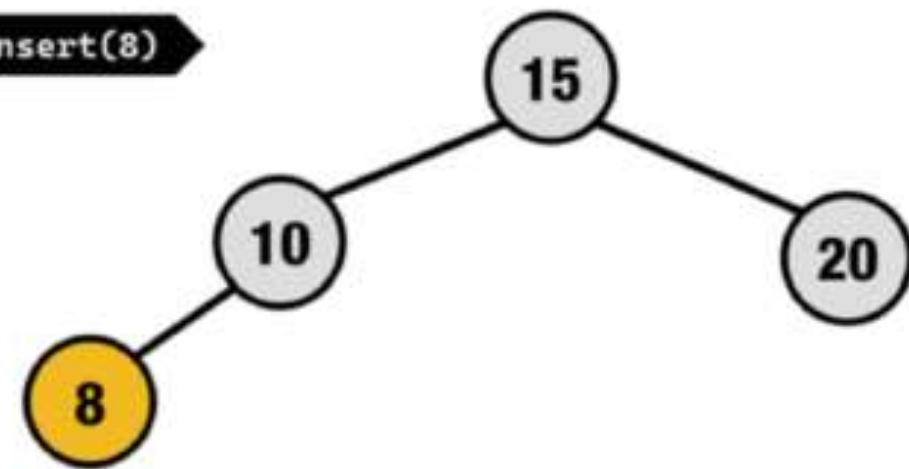
Insert(10)



Insert(20)



Insert(8)



Insert(12,17,25)

